

Contrastive domain generalization convolution neural network correcting the drift of gas sensors

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ABSTRACT

The drift problems of gas sensors caused by aging ingredients and unexpected environmental impacts, which can severely decrease the performance and service life of electronic noses, have been widely discussed. Many scholars believe that the essential reason for sensor drift can be attributed to the different data distributions in the latent feature space. However, traditional drift compensation methods like OSC, GasNet, SniffMultinose, and SniffConv are costly and complex. This paper has introduced an algorithm called contrastive domain generalization convolution neural network (CDCNN) to resolve this problem. For the first time, domain generalization and contrastive learning were used as the drift compensation methods for the gas sensor. A novel data augmentation was proposed to enrich the datasets. Feature generation module is used to simulate the drift of gas sensors. Contrastive learning adapts to the unseen areas in the latent feature space. The artificially generated features and the original features are drawn closer in the feature space to improve the algorithm's generalization ability. Experiments on long-term drift data show that CDCNN achieves high accuracy (0.7230). The experimental results show that the CDCNN algorithm is more suitable for practical applications due to less resource consumption and looser constraints.

1. Introduction

The electronic nose (E-nose) is an artificial olfaction technique that combines the gas sensor array with the recognition algorithms, and it has apparent advantages in fast speed and high accuracy. With the development of advanced sensitive materials and artificial intelligence algorithms, E-nose has been widely applied in many fields [1], such as environment monitoring [2], food inspection [3], and power system diagnosis [4,5].

However, due to aging and unpredictable environmental influences, sensor performance can degrade over time. This leads to sensor drift, where the same set of sensors may produce different response values for the same gas concentration. Scholars attribute sensor drift to aging, poisoning, and environmental fluctuation. However, the mechanism behind sensor drift is complex, making it hard to describe and compensate for. To improve an E-nose's detection performance and lifespan, drift compensation approaches must be explored.

Over the past twenty years, researchers have been working to solve

the drift problems of gas sensors. According to previous studies, the existing methods to solve the drift problems can be roughly divided into traditional machine learning and novel deep learning. In the preliminary stage of machine learning, most scholars relied on the assumption that the observed drift presents a reference direction rather than a random distribution. For instance, Artursson et al. proposed a method based on PCA and PLS to eliminate the effects of drift [6], but these methods improved the performance of e-nose at the expense of long-term robustness. Because the sensitivity characteristics of the sensors change in a complex way over time, the classification model needs to correct new calibration samples periodically. Other researchers disagree with this hypothesis. Alexander Vergara and Vembu et al. collected sensor data from long-term detections and used ensemble learning to compensate for drift. [7], M. Padilla used Orthogonal Signal Correction (OSC) to find the undesired element by calculating a subspace orthogonal to the label space [8]. The ELM is currently receiving widespread attention. S. Lu et al. proposed a domain transformation semi-supervised weighted kernel extreme learning machine (DTSWKELM) algorithm [9].

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This algorithm achieves high recognition accuracy under semi-supervised conditions. J. Yan et al. proposed a new algorithm for subspace alignment extreme learning machine (SAELM) based on the method of subspace alignment [10].

With the development of deep learning, the hybrid deep neural networks with Multilayer Perceptron are applied on the E-nose to solve the drift problems. For example, Liu et al. proposed algorithms (SSAE and DBNs) that compensate for the gas sensor drift by deep learning [11]. These algorithms use a fully connected layer, and the recognition accuracy is not high. It is mentioned in the article that these methods have a correct rate of only 0.65 for identifying drift data. Nevertheless, all the above methods require the acquisition of samples and their associated labels for drift calibration, which is impossible under non-laboratory conditions. As a result, transfer learning is proposed as a less expensive alternative. Liu Bei et al. have attempted to use broad learning and maximum mean distance (MMD) [12] to withstand the effect of sensor drift. Online learning combined with transfer learning is more suitable for real applications [13]. Even though transfer learning only requires a few samples, it still needs to visit the target domain, which is almost impossible in actual scenarios. Y. Zhang et al. have proposed a Target-domain-free domain adaptation convolutional neural network (TDACNN) [14]. The network does not need to access the target domain data during training, making it more suitable for practical applications. However, the source domain data must be revisited when testing, which creates new problems. For instance, the network will fail if the source domain data is damaged. Additionally, accessing the source domain data consumes more resources and is inconvenient for deployment.

To address the drift issue, we have proposed a novel algorithm named the contrastive domain generalization convolution neural network (CDCNN). It combines domain generalization and contrastive learning to solve the sensor drift problem. The dataset without the drift effect is defined as the source domain, and the dataset with the drift

effect is defined as the target domain. It is noted that the proposed CDCNN is target-free, and only the source domain is required. The framework of the CDCNN includes data manipulation, domain generalization, and contrastive learning. Data manipulation enriches input information and extends the latent feature space. Domain generalization improves the network's ability to fit the target domain, while contrastive learning minimizes the distance of global samples with the same label. The experimental results show that the CDCNN algorithm is more suitable for practical applications due to less resource consumption and looser constraints.

2. Methodology

As mentioned above, the CDCNN algorithm compensates for sensor drift from the perspective of domain generalization and contrastive learning, which makes target-free come true. In this section, the details of the CDCNN algorithm are introduced.

2.1. The introduction of domain generalization and contrastive learning

In scenarios where image recognition must be performed on out-of-distribution (OOD) data, domain generalization is a standard technology that can be used to overcome the limitations of deep learning. As illustrated in Fig. 1(a) and Fig. 1(b), domain generalization enables the use of different distributions from the training dataset. Existing domain generalization methods can be divided into three categories: data manipulation, representation learning, and learning strategy. Data augmentation and data generation make up data manipulation, which can get more information from artificial domains to improve performance in the target domain. However, this approach may decrease performance in unvisited domains due to the random distribution of the generated data in the feature space. Representation learning typically contains domain-invariant representation learning and feature

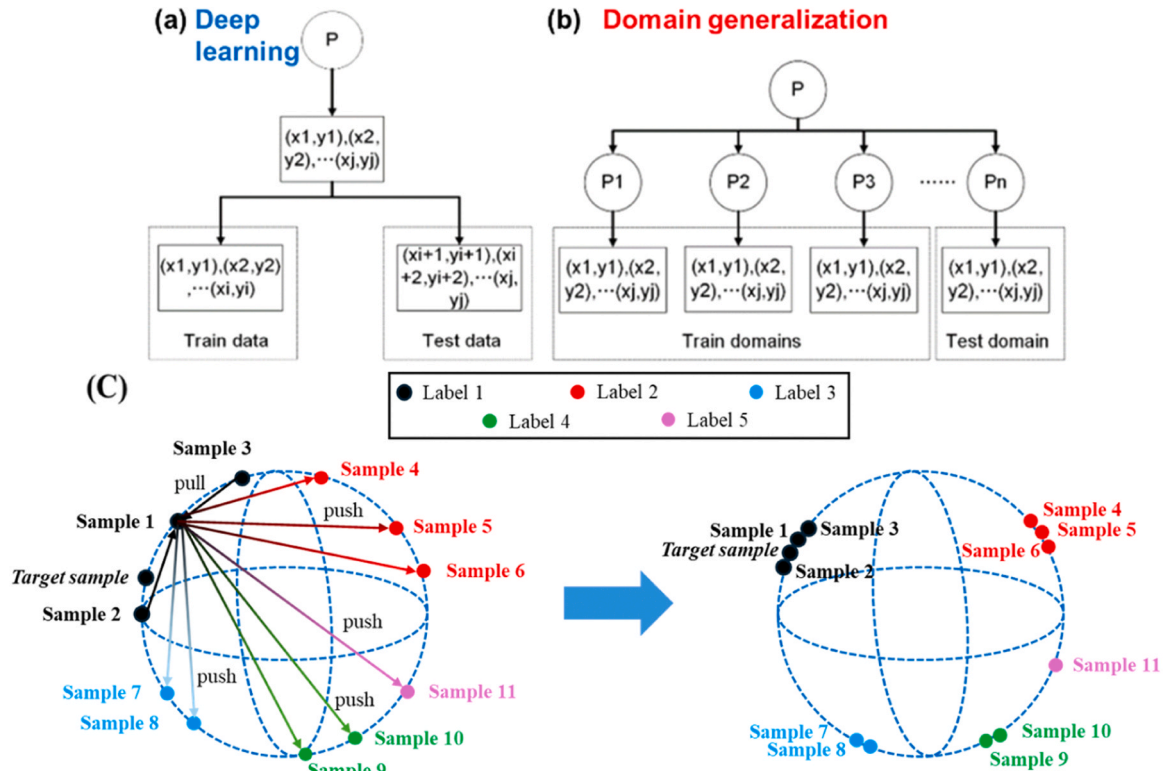


Fig. 1. (a) is the illustration of deep learning. (b) is the illustration of domain generalization. P is the distribution of the dataset. (c) is the illustration of contrastive learning.

disentanglement, which aim to encourage the network by extracting specific features. However, this approach lacks a theoretical foundation for decomposing domain-irrelevant features. The learning strategy category includes ensemble learning, meta-learning, and gradient operation. While effective, this method is associated with high costs and lower accuracy than other methods, making it a less popular alternative. The CDCNN consists of an innovative data augmentation method and a feature generation method that belongs to data manipulation. The algorithm optimizes the unvisited latent space in data manipulation through contrastive learning.

The purpose of contrastive learning is to minimize the intra-class distance and maximize the inter-class distance within the unit range. Fig. 1(c) sketches the theory of contrastive learning. Momentum Contrast builds a large and consistent dictionary on-the-fly that facilitates contrastive unsupervised learning [15]. Chen et al. illustrated a simple framework for contrastive learning of visual representations (SimCLR)[16]. They use projection heads to achieve SOTA in various tasks. While contrastive learning is self-supervised, it can also be adapted to a fully-supervised setting [17]. In the CDCNN algorithm, dot-similarity is used to measure the distance between different samples, and local intra-class distances are minimized with global intra-class distances to improve the accuracy of data processing for unvisited domains.

To explain the importance of contrastive learning, we compared CDCNN with CDCNN without contrastive learning (CDWC), as shown in Supporting Information (S2). The accuracy in the testing set decreases when the contrastive learning block is removed from CDCNN, as demonstrated in Fig. S1. The impact of batch size on accuracy is illustrated in Fig. S2. These results show that contrastive learning can enhance the accuracy of the drift compensation neural network. The confusion matrix in Fig. S3 for CDCNN and CDWC indicates that contrastive learning leads to higher recognition accuracy for different gases.

2.2. The overall introduction of the CDCNN algorithm

The framework of the CDCNN algorithm is illustrated in Fig. 2, and the corresponding pseudo-code is presented in Table 1. Herein, Conv means the convolution layer. 3 Conv 128 means the kernel size is 3, and the number of filters is 128. The activation function chooses RULE. The loss function of the algorithm is shown first, followed by details of the data augmentation and feature generation blocks. Additionally, the normalization signals of gas sensors are utilized to expand the original dataset through data augmentation blocks. The convolution neural network extracts the features of all the augmented data and the classifier, and the fully connected layer predicts the label of the data. The corresponding symbols of the dataset are defined as follows:

$$D = \{D_1, D_2, D_3, \dots, D_J, \dots, D_K\} \quad (1)$$

$$D_J = \{(X_1^J, Y_1^J), (X_2^J, Y_2^J), \dots, (X_N^J, Y_N^J)\} \quad (2)$$

$$X_i^J = [x_1^j, x_2^j, x_3^j, \dots, x_{ni}^j] \quad (3)$$

where X_i^J is the signal of the gas sensor and Y_i^J is the corresponding label of it. It contains ni sampling points x_i^j . X_i^J is sampled from D_J . D is the set of all different domains.

2.3. The loss function of CDCNN

The CDCNN algorithm is designed to train a model using data free from drift to identify data that has experienced drift accurately. To achieve this goal, we use domain generalization to improve the model's ability to generalize and contrastive learning to enhance the accuracy of the model's predictions on drift datasets. The parameters of CDCNN are upgraded as follows:

$$\min(L_{ce} + \lambda_{MSE} L_{MSE} + \lambda_{con} L_{con}) \quad (4)$$

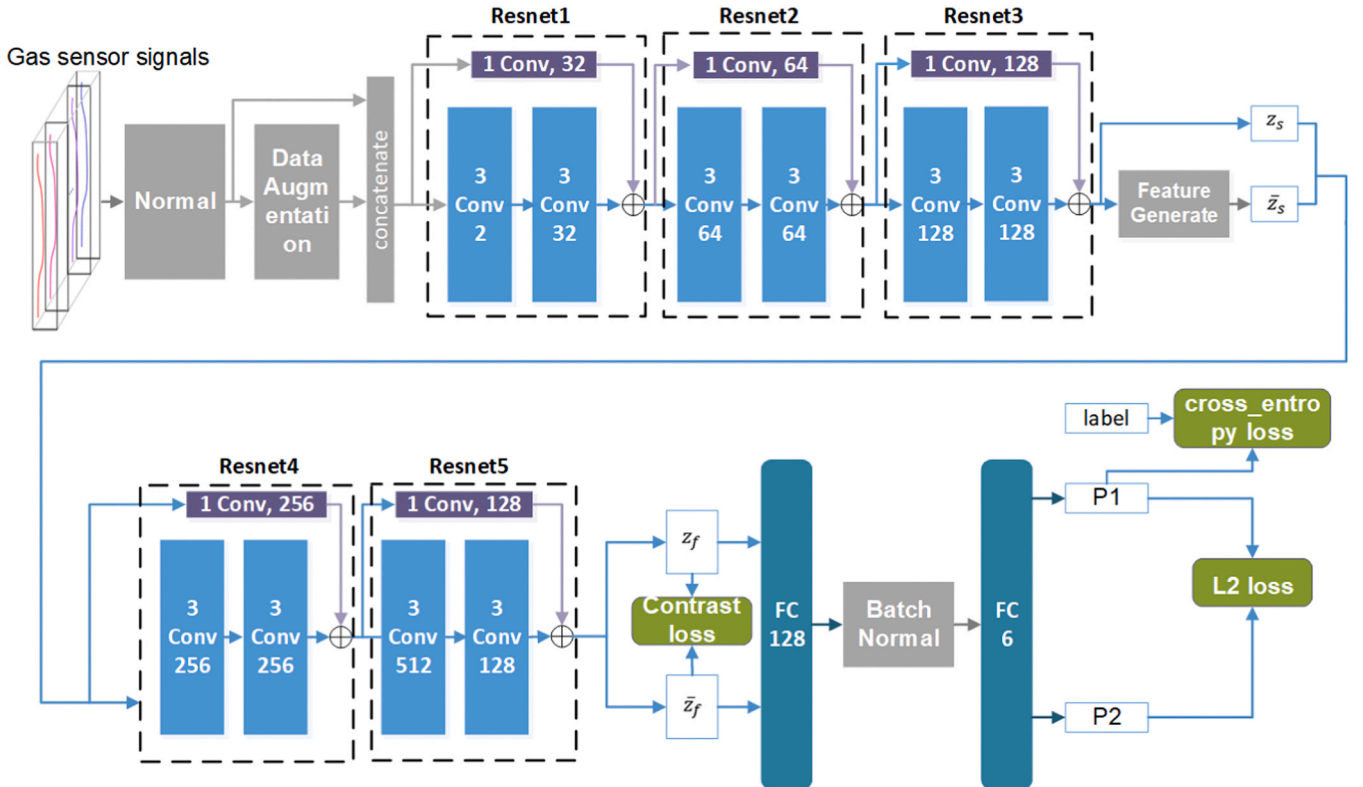


Fig. 2. The framework of CDCNN.

Table 1

The pseudo-code of the contrastive domain generalization convolution neural network.

Input: $X = [X_1^i, X_2^i, X_3^i, \dots, X_N^i]$, $Y = [Y_1^i, Y_2^i, Y_3^i, \dots, Y_N^i]$, initialized parameters R_s, R_f and FC	Output: learned parameters R_s, R_f and FC
1: Augment the input data $\bar{X} = \{X, A(X)\}$, $\bar{Y} = \{Y, Y\}$ 2: For $\bar{x}, \bar{y}$ in X, Y :	
3: Calculate the preliminary feature $Z_s = R_s(\bar{x})$ 4: Generate the artificial unseen domain feature $\bar{Z}_s = S(Z_s)$ 5: Calculate the finally feature $Z_f = R_f(Z_s)$, $\bar{Z}_f = R_f(\bar{Z}_s)$ and get Contrastive loss L_{con} 6: Compute the output of network $out = FC(Z_s)$, $out = FC(\bar{Z}_s)$ and calculate L_{l2} and L_{ce} 7: Calculate the weighted loss $L = L_{ce} + \lambda_{l2}L_{l2} + \lambda_{con}L_{con}$ 8: Compute the gradient of L, R_s, R_f and FC and then do the update	
9: end for	

where λ_{MSE} and λ_{con} are weighting factors that belong to $[0,1]$. The cross-entropy loss L_{ce} is the primary loss in our net for improving the identification ability of the gas signal of this neural network. L_{MSE} and L_{con} are auxiliary loss functions for encouraging the generalization ability. They represent MSE loss and L2 loss, respectively. The details of the loss function are talked about in [Supporting Information \(S1\)](#).

During forecasting, the program separates the data manipulation block from the prediction process, ensuring that the outcome is solely based on the input data.

2.4. Data manipulation

Data manipulation is an intuitive way to improve a network's generalization ability by visiting more samples. It consists of two parts, including data augmentation and feature generation. Data augmentation introduces diverse data, while feature generation generates artificial features from other domains. Some previous studies have transferred the gas signals into gray images and used two-dimensional convolution [18]. Instead, this work has designed the single-dimensional convolution [19] for less cost, as shown in [Fig. S4 \(a\)](#). Five ResNet layers are used to extract the features from the original data. The feature generates a block in the middle of the third and the fourth ResNet block.

3. Data augmentation

Data augmentation is a standard method for improving experiment accuracy without altering the network structure. However, it is challenging to augment gas sensor data using picture-enhanced tricks due to the character of the time series.

Gaussian white noise and constant noise are commonly used in gas detection projects, but they cannot accurately simulate the complex variations of gas sensors in real-world scenarios. To address this issue, we proposed a novel gas sensor signal augmentation approach inspired by Variational Autoencoders (VAE) shown in [Fig. S5](#) and verified this method in [Section 3](#).

VAE was proposed by Diederik P Kingma et al. to create nonexistent data from the original [20]. We use VAE theory to optimize the conventional signal augmentation approach. The mean μ_j and variance δ_j of the dataset can be extracted from the gas sensor signal as follows:

$$\mu_j = \frac{1}{n} \sum_{i=1}^n x_i \quad (5)$$

$$\delta_j = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_j)^2 \quad (6)$$

where j represents a set signal of the gas sensor and i means the data point in every sensor informal.

$\lambda * \mu_j + (1 - \lambda) * \mu_k$ and $\lambda * \delta_j + (1 - \lambda) * \delta_k$ are set as the mean and variance of Gaussian distribution $N(\lambda * \mu_j + (1 - \lambda) * \mu_k, \lambda * \delta_j + (1 - \lambda) * \delta_k)$, respectively. k means another gas sensor data different from j in this dataset. λ is a weight that varies from 0 to 1. The latent variable to augment data can be sampled from this Gaussian distribution. A latent variable is added to the original data to form a novel signal with the label changeless. If X represents the original signal and G represents the

augmentation data, the approach can be illustrated as follows:

$$G_j = f(N(\lambda * \mu_j + (1 - \lambda) * \mu_k, \lambda * \delta_j + (1 - \lambda) * \delta_k)) + X_j \quad (7)$$

where $f(\cdot)$ denotes the sampling data from the Gaussian distribution, the Gaussian distribution of each sample is extracted, and the parameters of the different distributions are mixed to create another Gaussian distribution. The noise used to augment the data is sampled from this newly generated distribution.

4. Feature generation

Data augmentation has effectively extended data but does not provide additional domain information. Thus, this algorithm proposes the feature generation block to generate unseen domain information to resolve this problem. The structure of the feature generation block is shown in [Fig. S4 \(b\)](#).

Gas sensor features, $z_s = R_s(X)$ $X \in \mathbb{R}^{B \times H \times C}$, extracted from original data are decomposed into high-frequency and low-frequency components in the feature generation block. $R_s(\cdot)$ means the first three convolutional layers; the remaining part is recorded as $R_f(\cdot)$. The features decomposition process is formulated as follows:

$$z_s^H = \text{sample}(\text{MaxPool}(z_s)) \quad (8)$$

$$z_s^L = z_s - z_s^H \quad (9)$$

where "MaxPool" denotes the maximal pooling operation to extract the represented value of the feature, and "sample" indicates the nearest neighbor up-sampling operation. Since the low-frequency reserves only some feature correlation domain label information, the network normalizes original data use μ_s^L and δ_s^L which are obtained as follows:

$$\mu_s^L = \frac{1}{C_s} \sum_{m=1}^{C_s} \frac{1}{H_s} \sum_{n=1}^{H_s} z_{s,m,n}^L \quad (10)$$

$$\delta_s^L = \frac{1}{C_s} \sum_{m=1}^{C_s} \frac{1}{H_s} \sum_{n=1}^{H_s} (z_{s,m,n}^L - \mu_s^L)^2 \quad (11)$$

Note that these parameters $\mu_s^L, \delta_s^L \in \mathbb{R}^{B \times CH}$ denote the mean and variance of feature style, respectively. B and CH are batch size and the product of C , the channels of features, and H , the dimensionality of each feature, respectively. As we know, the low-frequency feature's parameters are related to the domain characteristics. The network changes the signal of the gas sensor to become another domain signal by changing these parameters.

For the generation of the invisible domain features, the batch-wise mean μ_s^L and variance δ_s^L of low-frequency features are calculated as follows,

$$\hat{\mu}_s^L = \frac{1}{B_s} \sum_{b=1}^{B_s} \mu_{b,s}^L, \hat{\delta}_s^L = \frac{1}{B_s} \sum_{b=1}^{B_s} (\mu_{b,s}^L - \hat{\mu}_s^L)^2 \quad (12)$$

$$\tilde{\mu}_s^L = \frac{1}{B_s} \sum_{b=1}^{B_s} \delta_{b,s}^L, \tilde{\delta}_s^L = \frac{1}{B_s} \sum_{b=1}^{B_s} (\delta_{b,s}^L - \tilde{\mu}_s^L)^2 \quad (13)$$

where $\hat{\mu}_s^L$ and $\hat{\delta}_s^L$ denote the statistics of the mean of domain distribution,

then $\tilde{\mu}_s^L$ and $\tilde{\delta}_s^L$ represent the statistics of variance of domain distribution. The $\mu_{b,s}^L$ and $\delta_{b,s}^L$ represent the mean and variance of each batch, respectively. B_s is the number of batches of z_s . We assume the mean and variance of the feature obey Gaussian. The sample features of another domain from Gaussian are as follows.

$$\mu_s' \sim N(\tilde{\mu}_s^L, \tilde{\delta}_s^L), \delta_s' \sim N(\tilde{\mu}_s^L, \tilde{\delta}_s^L) \quad (14)$$

Lastly, the signals belonging to the unseen domain are generated from the function illustrated in Eq. (15).

$$\bar{z}_s = z_s^H + \bar{z}_s^L \quad (15)$$

where $\bar{z}_s^L$ is a low-frequency feature of the new domain got from Eq. (16) as follows:

$$\bar{z}_s^L = \delta_s' \left(\frac{z_s^L - \mu_s^L}{\delta_s^L} \right) + \mu_s' \quad (16)$$

5. Results and discussion

5.1. Dataset description

To verify the compensation ability of the proposed algorithm, the authors evaluate its effectiveness on a long-term sensor drift dataset. This section introduces the details of this dataset. Several researchers have studied this dataset [13]. However, to our knowledge, authors are the first to compensate for drift with domain generalization and contrastive learning. This dataset comprises an E-nose array with 16 gas sensors. Everyone can visit it on the UCI dataset [7]. These gas sensors are four replicas of 4 different commercial Metal Oxide Semiconductor Gas Sensors (TGS2600, TGS2602, TGS2610, and TGS2620), which are produced by Figaro. This dataset was collected over three years from January 2008 to February 2011 with 36 months consisting of six kinds of gas signals (ammonia, acetaldehyde, acetone ethylene, ethanol, and toluene) at various concentration levels (5–1000 ppm), summing up to 13,910 samples in total. The authors of this dataset divide the data into ten batches according to the time interval collected. The details of the datasets are listed in Table 2.

Using the exponential moving average method, the authors have extracted eight features for each sample, including two steady-state features and six transient features. The two steady-state features are the difference and the ratio of the maximal resistance and the baseline value, respectively. The other six transient features reflect the sensor resistance dynamics of the increasing and decaying. These features comprise three values from the pre-recorded rising portion of the sensor response and three additional values for the decaying portion with 0.1, 0.01, and 0.001 smoothing parameters. Therefore, the dataset sample is a 128-dimensional (8×16) time series vector. To view the drift of the feature distribution of the dataset over time, we reduce the dataset dimensionality using PCA and map it to the two-dimensional principal item subspace. The mapped subspaces of each batch are shown in Fig. 3. The distribution of the first two main item plots drawn by different

batches has noticeable differences over time. The main reason for this phenomenon is the drastic gas sensor drift. The diverse color dots represent different gas kinds; the point in this picture is the gas sample.

5.2. Predicted results of the CDCNN algorithm

This section presents the network's experimental results, demonstrating our algorithm's feasibility, superiority, and practicality. As discussed, the dataset used in this experiment includes two steady-state features and six transient features of the original signal. A deep network to extract these features is not theoretically feasible. Therefore, we replaced it with a simple net. At the start of this algorithm, the normalization block is used to accelerate the convergence of the network. To get the core features of the gas signal, we use a neural network with five ResNet blocks. The kernel's shape of each ResNet is 3×1 , and there are two convolution layers and a shortcut function with 1×1 kernel in each block, and the channels of convolution increase from 1 to 128 step by step. Due to the simple net, to get enough features to forecast data labels, this network doesn't apply the pooling layer. In practice, we use the standard Stochastic gradient descent (SGD) and learning rate damping for higher accuracy.

According to previous research [32], the limitations of the used dataset include the non-random order of gas measurements, drift in baseline over time, and the spatial distribution of baseline variations. To avoid the issues that cause overestimating the accuracy, the first batch is employed as a source domain to train the algorithm, and the other batches that cannot be visited as the target domain, as shown in Fig. S7. So, the trained models can't acquire the baseline information from the other batches, and it makes the identification results reliable. The difference in collected time generates the main discrepancy between the source and target domains. Suppose the algorithm achieves high accuracy in the testing set after training on the first batch; it can be generalized to any other batches because the target domains can be considered random sampling from a time scale.

To study the feasibility of the data augmentation module in CDCNN, we compared it with traditional gas signal processing methods (Gaussian white noise (30 dB) and Constant noise (1)). We use these methods to process the raw data and project the results into a 3D principal component space through PCA. Compared with the other two conventional methods, the CDCNN data augmentation produced a massive difference from the raw data. In Fig. 4, the predicted accuracy of the four methods is 0.67, 0.70, 0.69, and 0.72, respectively. It proves that the data augmentation methods can effectively expand the raw data.

The feature generation block is the core of our CDCNN algorithm, which extends the existing features to simulate the drift of gas sensors in latent feature space. In this section, we try to study the effectiveness of the feature generation block. We use this module to generate additional features, which we then map onto the 2D principal component space through PCA, along with raw features. Fig. 5 shows the distribution of the generated features in space, where the blue dots represent the original signals, and the orange dots represent artificial features. Except

Table 2
The details of the dataset.

ID	month	Ethanol	Ethylene	Ammonia	Acetaldehyde	Acetone	Toluene	Total
Batch1	1,2	83	30	70	98	90	74	445
Batch2	3,4,8,9,10	100	109	532	334	164	5	1244
Batch3	11,12,13	216	240	275	490	365	0	1586
Batch4	14,15	12	30	12	43	64	0	161
Batch5	16	20	46	63	40	20	0	191
Batch6	17,18,19,20	110	29	606	574	514	467	2300
Batch7	21	360	745	630	662	649	568	3613
Batch8	22,23	40	33	143	30	30	18	294
Batch9	24,30	100	75	78	55	61	101	470
Batch10	36	600	600	600	600	600	600	3600

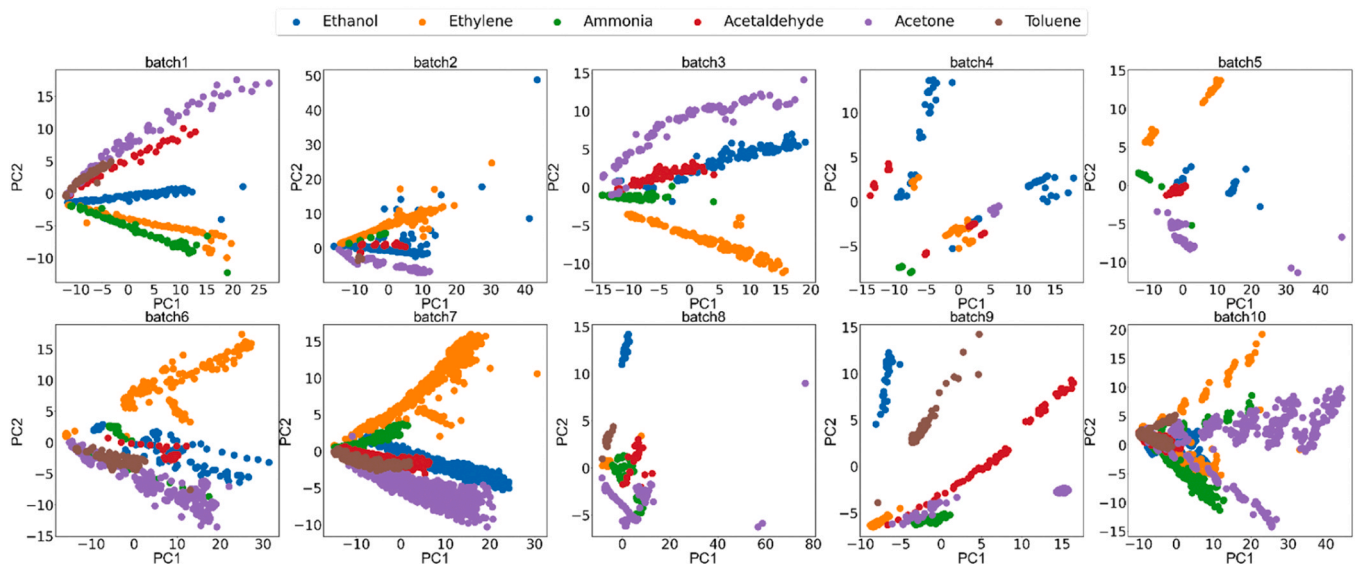


Fig. 3. Principal component (two-dimensional sub-space) image of drift dataset.

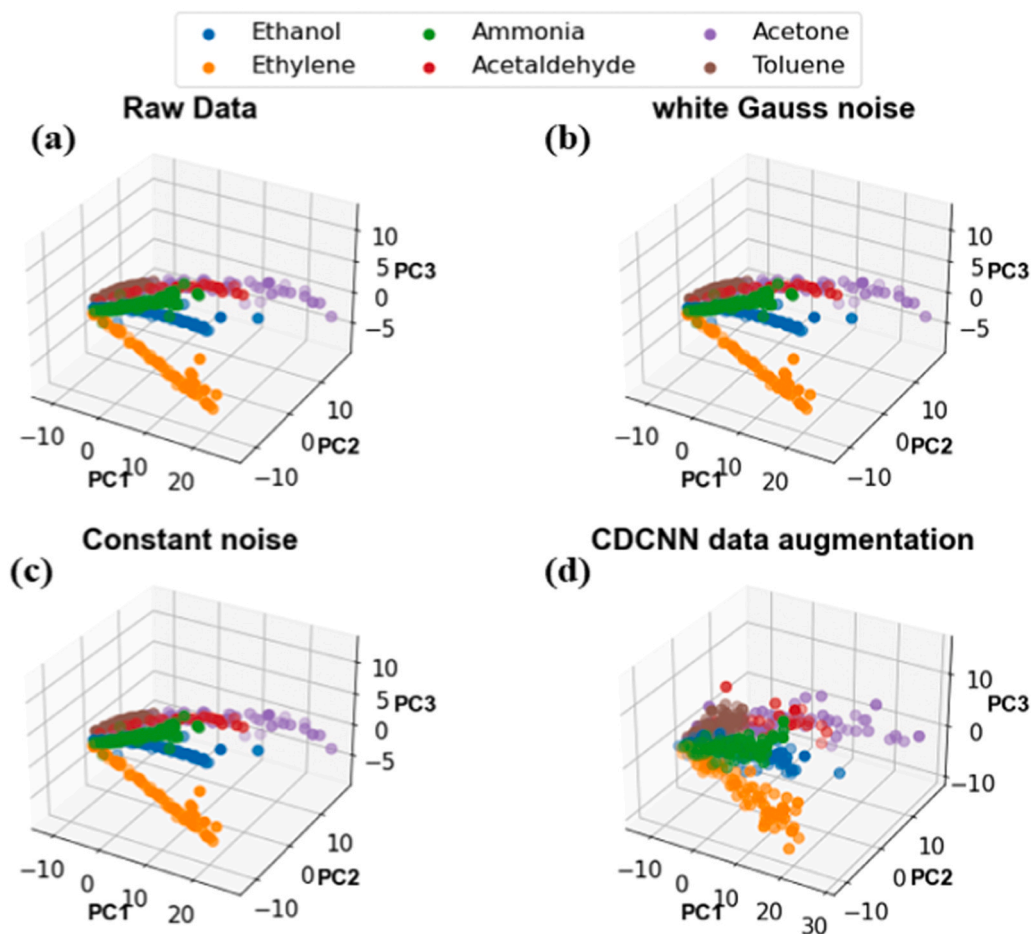


Fig. 4. PCA projections of (a) original data, and the data augmented by (b) white Gauss noise, (c) Constant noise, (d) CDCNN, respectively.

for the overlap in the characteristic distribution of ethanol, the generation characteristics of other gases are quite different from the original characteristics. Also, the distribution of the generated features in space is more dispersed, covering a wider spatial range, which helps to represent the drift effect better. The differences between the dots

confirm that the feature generation block successfully simulates the gas sensor's drift in latent feature space. Additionally, the PCA results of all batches after CDCNN processing have been provided in Fig. S8 of Supporting Information. In PCA space, the samples with the same labels are drawn closer to each other, while the samples with different labels are

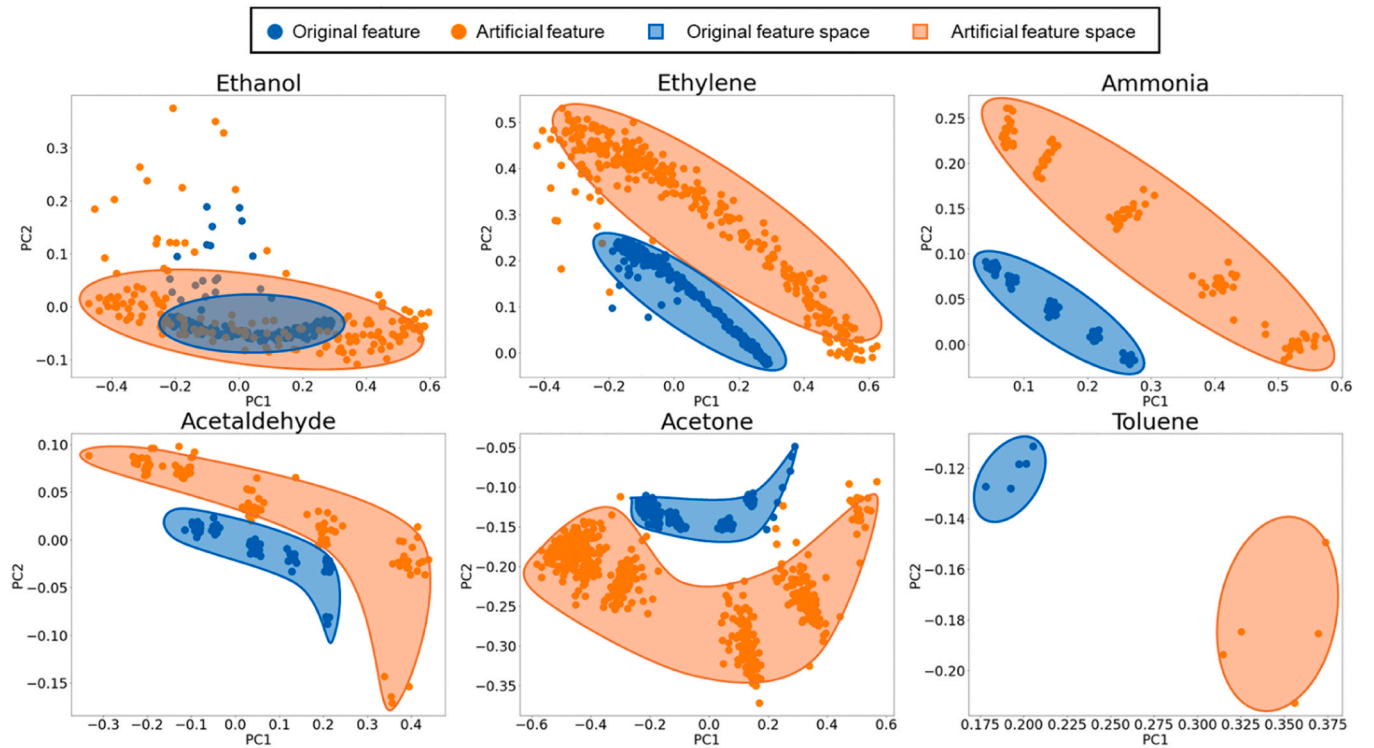


Fig. 5. The outcome of the feature generation block in one of the test domains (batch2). The blue dots in this picture are the original feature of the data, and the orange dots are the artificial feature to be used to simulate the drifts, which can appear in some possibilities.

pushed farther. The samples processed by CDCNN exhibit more pronounced clustering in comparison with the original data. It demonstrates that contrastive learning can improve the issue of sensor drift.

The first domain is used as a training set, and the others are used as test sets. Table 3 has performed the accuracy of CDWC, LDA-SVM, PCA-SVM, TDACNN, ResNet, SVM-RBF, SVM-GFK, DBNs [11], and as-designed CDCNN. It clearly shows that the CDCNN achieves the highest average accuracy; thus, this algorithm's advancement is further proof. The experiment in this article is rigorous, so many other approaches are applied to deal with this drift problem as a control experiment in the same setting. These approaches are composed of VGG19 [21], VGG16 [21], VGG13 [21], AlexNet [22], LeNet [23], GoogLeNet [24], ResNet (the backbone of CDCNN) [25], SniffMultinose [26], SniffConv [26], ODCNN [27], GasNet [28] and SVM-RBF, linear discriminant analysis (LDA), component correction based on PCA (CC-PCA) [6], PCA, locality preserving projections (LPP), direct standardization (DS) [29], generalized most minor squares weighting (GLSW) [30], Orthogonal signal correction (OSC) [8], TDACNN [14], CDWC. The score of these approaches is shown in Fig. 6. The result is consistent with the theory and shows that CDCNN can achieve a higher accuracy (0.7230) than the TDACNN (0.7221), and CDCNN is more suitable for practical use. Since TDACNN is trained in the training

domain, it continuously visits the source domain during testing. It uses ensemble learning (four classifiers) to predict the sample labels, while CDCNN only needs to visit the training domain once during training; without ensemble learning, only one classifier is required. The classifier of the TDACNN algorithm contains 54 K parameters, while the classifier of the CDCNN algorithm requires only 0.77 K parameters. Besides, we have added an experiment that uses ith batch of data to test all subsequent batches. The results in Supporting Information (S4) indicate the CDCNN model is not overfitting, and thus, the CDCNN model indeed has the ability to solve the issue of sensor drift.

6. Conclusions

In this paper, a novel algorithm named CDCNN was proposed to compensate for the drift of the gas sensor. Due to the unpredictable properties of sensor drift, it is expensive and complicated to calibrate the gas sensor. However, CDCNN can overcome the drift effect because of its robustness and generalization ability. Furthermore, data augmentation and feature generation methods were utilized to enrich the dataset, which helps to simulate sensor drift in more diverse domains. At the same time, it is proved that our method can minimize the local distribution distance under the global distribution distance. Contrastive

Table 3
The accuracy of various algorithms for various domains.

Task	1–2	1–3	1–4	1–5	1–6	1–7	1–8	1–9	1–10	Average	Reference
SVM-RBF	0.7436	0.6103	0.5093	0.1827	0.2826	0.2881	0.2007	0.3426	0.3447	0.3894	[31]
LDA-SVM	0.4727	0.5776	0.5093	0.6244	0.4148	0.3742	0.6837	0.5234	0.3117	0.4991	[31]
PCA-SVM	0.8240	0.8480	0.8012	0.7513	0.7357	0.5616	0.4864	0.6745	0.4914	0.6860	[31]
SVM-GFK	0.7447	0.7015	0.5978	0.7509	0.7399	0.5459	0.5588	0.7023	0.4185	0.6400	[31]
TDACNN	0.8956	0.8383	0.7764	0.7563	0.7436	0.6208	0.7510	0.6085	0.5088	0.7221	[14]
DBNs	0.7846	0.8398	0.7391	0.7107	0.7057	0.5015	0.7449	0.6732	0.2994	0.6665	[11]
CDWC	0.7599	0.7721	0.7049	0.7891	0.8514	0.5775	0.5824	0.5684	0.4291	0.6705	This work
ResNet	0.7692	0.6618	0.6424	0.7160	0.7253	0.5024	0.6602	0.6035	0.4302	0.6344	This work
CDCNN	0.8271	0.7683	0.7244	0.7891	0.9318	0.6182	0.7152	0.5664	0.5663	0.7230	This work

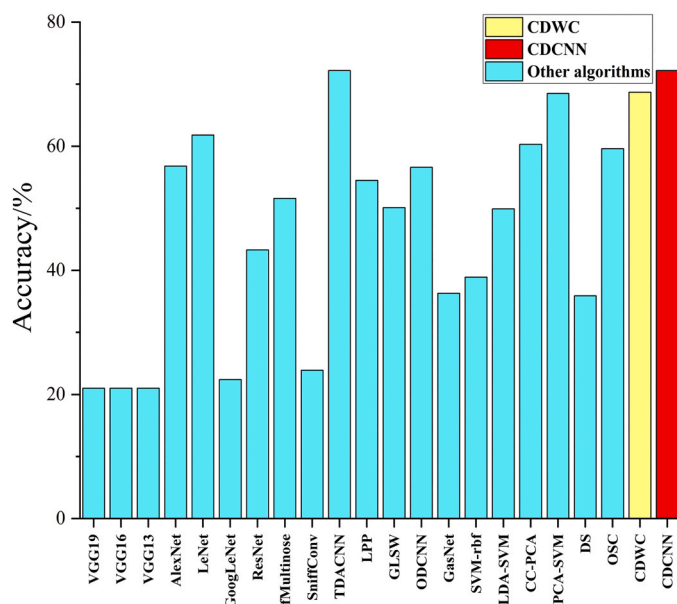


Fig. 6. The accuracy of different methods. The CDCNN algorithm can achieve the highest accuracy in these various methods, and the target domain is unnecessary.

learning ensures the robustness of the generalization algorithm and overcomes the drawbacks of data manipulation. Overall, the primary achievements in this work are as follows:

- (1) For the first time, a domain generalization method has been applied to compensate for the drift effect of the metal-semiconductor gas sensor, which was caused by the variation in distribution in feature space.
- (2) A novel data manipulation method, including data augmentation and feature generation, has been designed, and its effectiveness has been verified.
- (3) Combining contrastive learning with data manipulation, the CDCNN has an excellent effect on the drift compensation (0.7230) without visiting the target data, which is much higher than the other traditional methods.
- (4) Compared with the TDACNN algorithm, which also does not need to access the target domain, CDCNN consumes fewer resources during testing and has looser data constraints.

Although the domain generalization to solve the drift compensation has not attracted much attention, it is still a method with great potential. Further efforts should be made in data manipulation, network structure, and loss function to improve the accuracy of this algorithm.

CRedit authorship contribution statement

Jianbin Pan: Validation, Investigation. **Huan Yuan:** Writing – review & editing. **Xianbo Huang:** Software, Methodology, Data curation, Writing – review & editing. **Aijun Yang:** Project administration. **Mingzhe Rong:** Funding acquisition. **Xiaohua Wang:** Supervision. **Jifeng Chu:** Writing – original draft, Resources, Investigation, Conceptualization. **Renhong Yao:** Visualization, Software, Formal analysis, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data Availability

The data that has been used is confidential.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.sna.2024.115314](https://doi.org/10.1016/j.sna.2024.115314).

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